School of Computing Technologies

ISYS1055/1057/3412 (Practical) Database Concepts

Assessment 1: Database Design

Assessment type: PDF

Word limit: N/A

Draft Due Date: 13 August at 23:59 (Melbourne time) – Week 4 (otherwise 2 mark deduction)
Final Due Date: 27 August at 23:59 (Melbourne time) – Week 6

Weighting: 20%, 20 Marks

Overview

The objective of this assignment is to measure your understanding of the basic concepts in the relational database model and using entity-relationship model for database design. The assessment is in two parts, split into four tasks which cover Basic ER Modelling and Basic Relational Modelling. The tasks are as follows.

Part A: Entity-Relationship Modelling (12 Marks)

1. Design and plan for the implementation of a database system, diagramming the design to a high standard using UML notation through the diagramming tool Lucidchart.

2. Model the activities of an organisation and present the model as an Entity-Relationship (ER) diagram. Analyse this ER diagram, and possibly modify it, based on additional client requirements.

3. Map an ER diagram into a relational database schema, showing every step of the mapping.

Part B: Relational Database Model (8 Marks)

4. Answer a series of short questions about a Relational Database model.

To complete this assessment, you must be familiar with Lucidchart, which is covered during the Week 1-4 activities.

Assessment criteria

This assessment will measure your ability to:

- Describe various data modelling and database system technologies.
- Explain the main concepts for data modelling and characteristics of database systems.

Course learning outcomes

This assessment is relevant to the following course learning outcomes:

CLO1	Describe the underlying theoretical basis of the relational database model and apply the theories into practice.
CLO2	Explain the main concepts for data modelling and characteristics of database systems.
CLO3	Develop a sound database design using conceptual modelling mechanisms such as entity-relationship diagrams.
CLO4	Develop a database based on a sound database design.

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Part A: Entity-Relationship Modelling

Task 1: Designing an Entity-Relationship Model

Exercise Central Study

Happy Hotels (HH) is an Australian hotel chain. The following are the requirements for managing data about staff, customers and bookings for HH.

Hotel

HH is a chain of hotels located around Australia where customers can book rooms for accomodation. Each hotel has a distinct address composed of number, street, suburb and state. They also have a name. Each hotel consists of several rooms, each with a room number (eg 101, 102, 201 etc), and a capacity. Rooms contain several different ameneties such as a 'TV', 'Washing Machine' etc each with a longer description (eg '50" Flat screen digital with dvd player', '11Kg Washer' etc). Each room only has one of each type, but they can come in different forms (eg '75" Flat screen digital').

Staff

Staff are rostered on dates to maintain specific rooms of different hotels. The start and end time of each rostered date is recorded along with the hourly pay rate for the rostered hours. The rooms (and hotel) they are rostered for can change from day to day, and they could have more than one room rostered on the same day - but at different times (that cannot overlap), and with potentially different pay rates. The system records the unique employee id, the name and the role of the staff member (eg cleaner).

Booking

Customers can book rooms for accommodation. The start date and duration (in days) of the room booking, is recorded along with the name under which the booking is made. The system needs to record the status of the booking (eg reserved, checked-in, checked-out, cancelled etc)

Customer

Each Customer has a unique creditCardNo, with an expiry date, firstname and lastname.

Based on the given description, model the given business rules, and present your model as an Entity-Relationship (ER) diagram. Carefully state any assumptions that you make. In your ER diagram, you must properly denote all applicable concepts, including weak or strong entities, keys, composite or multi-valued attributes, relationships and their cardinality and participation constraints.

If you cannot represent any of this information in the ER model, clearly explain what limitations in the ER model restrict you from representing your model.

You **must** use UML notation and the diagramming tool Lucidchart to draw your diagram. Your diagram must be drawn to a high standard with minimal clutter. You are **not** required to map the ER model to relational model.

A special note: This is an open-ended question with many different models that can be derived. Your model is assessed based on how accurately it represents business rules described above.

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Task 2: Designing an Entity-Relationship Model

Part A: Initial Design

Bonza Books Case Study

Bonza Books is an Australian online book store.
You are asked to design a database for managing the book inventory and customer orders. Requirements for the database are as follows:

- There are many warehouses that stock books. Each warehouse has a unique number, and has an address.
- Books have an ISBN and a title. The system keeps track of the quantity of each book in the warehouse. Books can be stocked in more than one warehouse, and the system needs to track books that may be stocked in the future.
- Each order has an order number, a date and a (delivery) address. The order numbers are recycled each day and each order can contain many copies (quantity) of different books.
- Each book is published by a single publisher, each with a different name and with an address.

Based on the given description, model the business rules of Bonza Books, and present your model as an Entity-Relationship (ER) diagram. Carefully state any assumptions that you make. In your ER diagram, you must properly denote all applicable concepts, including weak or strong entities, keys, composite or multi-valued attributes, relationships and their cardinality and participation constraints.

If you cannot represent any of this information in the ER model, clearly explain what limitations in the ER model restrict you from representing your model. Avoid introducing unnecessary artificial keys.

You **must** use UML notation and the diagramming tool Lucidchart to draw your diagram. Your diagram must be drawn to a high standard with minimal clutter. You are **not** required to map the ER model to relational model.

A special note: This is an open-ended question with many different models that can be derived. Your model is assessed based on how accurately it represents business rules described above.

Part B: Client Adjustments

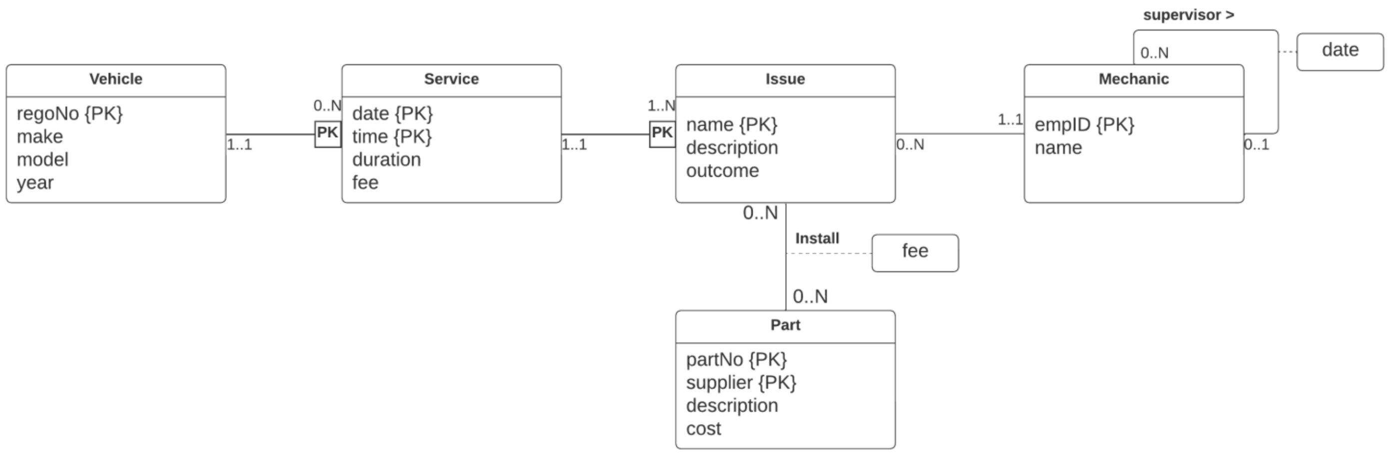
After presenting your ER model to Bonza Books management, you are asked if it can be used to perform the following additional tasks.

- Each order is processed by a particular staff member, with a name and a distinct staff number.
- Senior staff members supervise several more junior staff members, but the supervisor may change over time and the system must record the start and end date for each supervision period (including historic information).
- Staff work at a specific warehouse, and the start date is recorded.

For each one of the tasks specified above, explain how your ER diagram is supporting it. If it is not possible to achieve any of the tasks above given your current design, state why, modify the model, provide the modified ER diagram (in addition to your original ER diagram), and explain how the new model achieves the missing requirements.
As part of your discussion explain if (and why) your design can enforce the constraint that staff can only process orders for books from the warehouse they specifically work at. If this cannot be enforced, again explain why not.

Task 3: Mapping an ER Model to a Relational Database Schema

Consider the following ER diagram, which shows aspects of a vehicle repair centre.



You are requested to map the above ER diagram into a relational database schema. Show every step of the mapping. **No marks are awarded to the final schema if you do not show the partially built schema at the end of each step.** Clearly indicate the primary key (underlined) and foreign keys (with an asterisk) in each relation.

Part B

Task 4: Relational Database Model

This section contains the schema and a database instance for the Employee database that stores employee data for an organisation. The data includes items such as personal info (e.g., name, phone, salary), hierarchy (supervisors of employees), departments of the organisation (e.g., name and location of each department, who the manager is), jobs (e.g., titles, salary range), and a history for past contracts with each employee. A database instance is shown in Figure 2 followed by the database schema.

Employee:

employeeID	firstName	lastName	phoneNumber	hireDate	empJobID	salary	departmentID	superEmpID
50	Adam	Smith	1234	26/10/2009	22	$66,000	2	66
66	Tom	Moosa	1234	10/12/2016	10	$140,000	2	66
10	Jonny	Deans	1236	21/08/2002	33	$70,000	1	12
12	Adam	Jones	1247	08/08/2009	10	$138,000	1	12
18	Joseph	Ryan	1277	05/05/2020	10	$150,000	3	18

Department:

departmentID	departmentName	managerID	locationID
1	IT Services	12	10
2	Accounting	66	20
3	Human Resources	18	30

Job:

jobID	jobTitle	minSalary	maxSalary
10	Dept Manager	$120,000	$150,000
22	Accountant	$60,000	$80,000
33	Programmer	$60,000	$80,000
45	Senior Programmer	$70,000	$120,000

Location:

locationID	streetAddress	postalCode	City	stateProvince	countryID
10	123 Collins St	3000	Melbourne	VIC	1
20	222 Bourke St	3000	Melbourne	VIC	1
30	555 Swanston St	3000	Melbourne	VIC	1

Country:

countryID	countryName
1	Australia
2	Vietnam
3	Spain

JobHistory:

employeeID	startDate	endDate	jobID	departmentID
10	01/01/2001	10/04/2002	33	1
10	11/04/2002	20/08/2002	33	1
12	01/01/1998	05/10/2003	33	1
12	06/10/2003	06/10/2004	33	1
12	07/10/2004	07/08/2009	33	1
18	01/01/2019	04/05/2020	10	3

Figure 2: Employee Database Instance

The database schema is shown below, and the meaning of most attributes is self-explanatory. "superEmpID" is the employeeID of the supervisor of an employee. Primary keys are underlined, and foreign keys are annotated with a *.

Employee(<u>employeeID</u>, firstName, lastName, phoneNumber, hireDate, empJobID, salary, departmentID, superEmpID)

Department(<u>departmentID</u>, departmentName, managerID*, locationID*)

Job(<u>jobID</u>, jobTitle, minSalary, maxSalary)

Location(<u>locationID</u>, streetAddress, postalCode, city, stateProvince, countryID*)

Country(<u>countryID</u>, countryName)

JobHistory(<u>employeeID*, startDate, endDate</u>, jobID*, departmentID*)

The following table further clarifies the connection between the keys across multiple tables.

```
JobHistory.employeeID -----> Employee.employeeID
Department.managerID -----> Employee.employeeID
JobHistory.departmentID -----> Department.departmentID
JobHistory.jobID -----> Job.jobID
Location.countryID -----> Country.countryID
Department.locationID -----> Location.locationID
```

Figure 3: Keys

The following questions must be answered based on the given database schema and instance. Where explanation is required, each answer should be a SHORT passage of at most several lines.

QUESTION 4.1: Does the database schema ensure that there is a supervisor associated with each employee? Explain your answer.

Questions 4.2: What foreign key constraints are missing from Figure 3? Write down the missing constraints in the format shown in figure 3. Show the updated schema for the corresponding relations.

QUESTION 4.3: The 'Human Resource' department is being disbanded and removed from the database. All staff in this department will be terminated and also removed from the database.

The following SQL statements are intended to record all the changes required in the database instance. Will they work? If they are sufficient to achieve the requirements specified above, explicitly mention they are sufficient. If there are any shortcomings, identify them, how they can be rectified and briefly justify your answer.

```
DELETE FROM Department WHERE deparmentID = 3;
```

QUESTION 4.4: The employee named Jonny Deans changed his job to become a Senior Programmer on 07/07/2021. The following SQL statement intends to make the required changes to reflect Jonny's promotion.

```
UPDATE Employees SET empjob_id=45, hire_date='07/07/2020' WHERE phoneNumber=1236;
```

Explain if there any issues with the outcome of the update and how it should be fixed?. After running the above query, consider the request "find all the past contracts that Jonny Deans used to have". Can this request be completed using the given database schema and after the above statement is run? If yes, explain how the request can be answered. If no, explain what is missing and how it should be fixed.

QUESTION 4.5: Explain what the result of executing the following SQL statement on the database instance will be.

```
UPDATE Department SET departmentID=4 WHERE departmentID=1;
```

Identify all changes that must be completed to allow this to successfully execute.

QUESTION 4.6: Write an SQL statement to create the Location table including all the constraints, assuming all the tables that Location depends on already exist in the database. Make reasonable assumptions for the data type associated with each field. Your SQL statement must be valid for SQLite Studio environment and free of any errors.

QUESTION 4.7: Write an SQL statement to create the Country table including all the constraints, assuming all the tables that Country depends on already exist in the database. Make reasonable assumptions for the data type associated with each field. Your SQL statement must be valid for SQLite Studio environment and free of any errors.

QUESTION 4.8: On '01/01/2022' a new employee 'John Smith' is hired with the job title of CEO (job id 66) with a salary of $200,000, though the salary range could be $150,000 – $250,000. John's has the employee id 99, phone number 0123, he is assigned as the manager of the new department named 'Management' (department id 4), located at 123 Collins St Melbourne Victoria Australia. Futhermore, John is now the supervisor of each department manager. You are asked to update the given database instance so that it includes ALL relevant changes required to store this consistently across all realations. Your SQL statement must be valid for SQLite Studio environment, free of any errors, and compatible/consistent with existing data in the instance in Figure 2.

Submission format

You should submit one PDF document with all answers together. Do not submit Word files.

You must use Lucidchart to work on Part 1 of your assignment. You may use Word or any other word processor to compile your submission. Use section titles to indicate which question you are answering. At the end, convert your answer sheet into PDF format. Microsoft Word has the option of saving your document in PDF format. If the conversion option is not available on your system or word processor, there are free PDF converters online you can utilise (e.g., http://convertonlinefree.com/).

Academic integrity and plagiarism

Academic integrity is about honest presentation of your academic work. It means acknowledging the work of others while developing your own insights, knowledge, and ideas.

You should take extreme care that you have:

- Acknowledged words, data, diagrams, models, frameworks and/or ideas of others you have quoted (i.e., directly copied), summarised, paraphrased, discussed, or mentioned in your assessment through the appropriate referencing methods.
- Provided a reference list of the publication details so your reader can locate the source if necessary. This includes material taken from Internet sites.

If you do not acknowledge the sources of your material, you may be accused of plagiarism because you have passed off the work and ideas of another person without appropriate referencing, as if they were your own.

RMIT University treats plagiarism as a very serious offence constituting misconduct.

Plagiarism covers a variety of inappropriate behaviours, including:

- Failure to properly document a source
- Copyright material from the internet or databases
- Collusion between students

For further information on our policies and procedures, please refer to the University website.

Referencing guidelines

Where referencing is required, use RMIT Harvard referencing style for this assessment.

Refer to the RMIT Easy Cite referencing tool to see examples and tips on how to reference in the appropriated style. You can also refer to the library referencing page for more tools such as EndNote, referencing tutorials and referencing guides for printing.

Penalties for late submissions

Assignments received late and without prior extension approval or special consideration will be penalised by a deduction of 10% of the total score possible per calendar day late for that assessment.

Assessment declaration

When you submit work electronically, you agree to the assessment declaration.